

Math 526, F06
Quiz 2

Name:
SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$.

a) Find the $\mathbf{L} - \mathbf{U}$ factorization of $\mathbf{A}$.

Solution:

$$\begin{array}{cc} \left[\begin{array}{ccc} & & \end{array} \right] & \left[\begin{array}{ccc} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{array} \right] \\ \left[\begin{array}{ccc} & & \\ -2 & & \\ 3 & & \end{array} \right] & \left[\begin{array}{ccc} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{array} \right] \\ \left[\begin{array}{ccc} & & \\ -2 & & \\ 3 & 3 & \end{array} \right] & \left[\begin{array}{ccc} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{array} \right] \end{array}$$

So we have

$$\mathbf{L} = \begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & 3 & 1 & \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}.$$

b) Using the above factorization to solve $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Solution:

First step: Solve $\mathbf{Ly} = \mathbf{b}$.

$$\begin{bmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & 3 & 1 & \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$$

We get

$$y_1 = 2, \quad y_2 = 6, \quad y_3 = -20.$$

Second step: Solve $\mathbf{Ux} = \mathbf{y}$.

$$\begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ -20 \end{bmatrix}$$

We have

$$x_3 = -1, \quad x_2 = -\frac{1}{2}, \quad x_1 = \frac{1}{2}.$$

So we have

$$\mathbf{x} = \begin{bmatrix} 1/2 \\ -1/2 \\ -1 \end{bmatrix}.$$